Four Basic Counting Principles I

- A *partition* of a set S is a collection $S_1, S_2, \ldots, S_m$ of subsets of S such that each element of S is in exactly one of those subsets:

$$S = S_1 \cup S_2 \cup \cdots \cup S_m,$$
$$S_i \cap S_j = \emptyset, \quad (i \neq j).$$

The subsets $S_1, S_2, \ldots, S_m$ are called the *parts* of the partition. The *number of objects of a set* S is denoted by $|S|$ and is sometimes called the *size* (*cardinality*) of S.

Addition Principle.

Suppose that a set S is partitioned into parts $S_1, S_2, \ldots, S_m$. The number of objects in S can be determined by finding the number of objects in each of the parts, and adding the numbers so obtained:

$$|S| = |S_1| + |S_2| + \cdots + |S_m|.$$



Four Basic Counting Principles II

- The art of applying the addition principle is to partition the set S into "not too many manageable parts".

Example

Suppose we wish to find the number of different courses offered by the University of Science and Technology of China. We partition the courses according to the school in which they are listed. Provided there is no cross-listing, the number of courses offered by the University equals the sum of the number of courses offered by each department.

Example

A student wishes to take either a mathematics course or a biology course, but not both. If there are 4 mathematics courses and 3 biology courses for which the student has the necessary prerequisites, then the student can choose a course to take in $4 + 3 = 7$ ways.



Four Basic Counting Principles III

Multiplication Principle.

Let S be a set of ordered pairs (a, b) of objects, where the first object a comes from a set of size p, and for each choice of object a there are q choices for object b. Then the size of S is $p \times q$:

$$|S| = p \times q.$$

If a first task has p outcomes and, no matter what the outcome of the first task, a second task has q outcomes, then the two tasks performed consecutively have $p \times q$ outcomes.

Example

A student is to take two courses. The first meets at any one of 3 hours in the morning, and the second at any one of 4 hours in the afternoon. The number of schedules that are possible for the student is $3 \times 4 = 12$.



Four Basic Counting Principles IV

Example

Chalk comes in 3 different lengths, 8 different colors, and 4 different diameters. How many different kinds of chalk are there?

Solution

$3 \times 8 \times 4 = 96$.

Example

The number of ways a man, woman, boy, and girl can be selected from 5 men, 6 women, 2 boys, and 4 girls is $5 \times 6 \times 2 \times 4 = 240$.

Example

Determine the number of positive integers that are factors of the number $3^4 \times 5^2 \times 11^7 \times 13^8$.



Four Basic Counting Principles V

Solution

$5 \times 3 \times 8 \times 9 = 1080$.

Example

How many two-digit numbers have distinct and nonzero digits?

Solution

$9 \times 8 = 72$ or $90 - 9 - 9 = 72$.



Four Basic Counting Principles VI

Subtraction Principle.

Let A be a set and let U be a larger set containing A. Let

$$\overline{A} = \{x \in U : x \notin A\}$$

be the complement of A in U. Then the number $|A|$ of objects in A is given by the rule

$$|A| = |U| - |\overline{A}|.$$

Example

Computer passwords are to consist of a string of 6 symbols taken from the digits $0, 1, 2, \cdots, 9$ and the lowercase letters $a, b, c, \cdots, z$. How many computer passwords have a repeated symbol?



Four Basic Counting Principles VII

Solution

1. The number of all computer passwords is $|U| = 36^6 = 2,176,782,336$.
2. The number of passwords with no repeated symbol is $|\overline{A}| = 36 \cdot 35 \cdot 34 \cdot 33 \cdot 32 \cdot 31 = 1,402,410,240$.
3. The number of passwords with a repeated symbol is $|A| = |U| - |\overline{A}| = 774,372,096$.

Division Principle.

Let S be a finite set that is partitioned into k parts in such a way that each part contains the same number of objects. Then the number of parts in the partition is given by the rule

$$k = \frac{|S|}{\text{number of objects in a part}}.$$



Four Basic Counting Principles VIII

Example

There are 740 pigeons in a collection of pigeonholes. If each pigeonhole contains 5 pigeons, the number of pigeonholes equals $740/5 = 148$.

Example

You wish to give your Aunt Mollie a basket of fruit. In your refrigerator you have 6 oranges and 9 apples. The only requirement is that there must be at least one piece of fruit in the basket (that it, an empty basket of fruit is not allowed). How many different baskets of fruit are possible?



Four Basic Counting Principles IX

Solution

1. Ignore the requirement that the basket can not be empty and compensate for that later. $7 \times 10 - 1 = 69$.
2. Partition the non-empty baskets into two parts, S_1 and S_2, where S_1 consists of those baskets with no oranges and S_2 consists of those baskets with at least one orange. $|S_1| + |S_2| = 9 + 6 \times 10 = 69$.

- A great many counting problems are of the following types:
  - (i) Count the number of *ordered* arrangements or *ordered* selections of objects
    - (a) without repeating any object,
    - (b) with repetition of objects permitted (but perhaps limited).
  - (ii) Count the number of *unordered* arrangements or *unordered* selections of objects
    - (a) without repeating any object,
    - (b) with repetition of objects permitted (but perhaps limited).
- Arrangements or selections in (i) in which order is taken into consideration are called *permutations*, whereas arrangements or selections in (ii) in which order is irrelevant are called *combinations*.



Four Basic Counting Principles X

- Instead of distinguishing between nonrepetition and repetition of objects, it is sometimes more convenient to distinguish between selections from a set and a multiset. A *multiset* is like a set except that its members need not be distinct. For example, $M = \{a, a, a, b, c, c, d, d, d, d\}$, also denoted by $\{3 \cdot a, 1 \cdot b, 2 \cdot c, 4 \cdot d\}$. The numbers 3, 1, 2, and 4 are the *repetition numbers* of the multiset M. A repetition number of a set could be infinite, e.g., $\{\infty \cdot a, 2 \cdot b, \infty \cdot c, 4 \cdot d\}$.
- Thus a multiset breaks one of the cardinal rules of sets, namely, elements are not repeated in sets; they are either in the set or not in the set. The set $\{a, a, b\}$ is the same as the set $\{a, b\}$ but not so for multisets.

Example

How many odd numbers between 1000 and 9999 have distinct digits?



Four Basic Counting Principles XI

Solution

1. The units digit can be any one of 1, 3, 5, 7, 9.
2. The tens and hundreds digit can be any one of $0, 1, \ldots, 9$,
3. while the thousands digit can be any one of $1, 2, \ldots, 9$.
4. Thus there are 5 choices for the units digit.
5. Since the digits are to be distinct, we have 8 choices for the thousands digit, whatever the choice of the units digit.
6. There are 8 choices for the hundreds digit, whatever the first 2 choices were,
7. and 7 choices for the tens digit, whatever the first 3 choices were.
8. The answer $5 \times 8 \times 8 \times 7 = 2240$.

Example

How many integers strictly between 0 and 10,000 have exactly one digit equal to 5?



Four Basic Counting Principles XII

Solution

1. Let S be the set of integers in question. Partition S into the sets S_1, S_2, S_3, S_4 of one, two, three, and four-digit numbers in S.

$$|S_1| = 1, \qquad |S_2| = 8 + 9 = 17,$$
$$|S_3| = 8 \times 9 + 8 \times 9 + 9 \times 9 = 225,$$
$$|S_4| = 8 \times 9 \times 9 + 8 \times 9 \times 9 + 8 \times 9 \times 9 + 9 \times 9 \times 9 = 2,673.$$
$$|S| = 1 + 17 + 225 + 2,673 = 2,916.$$

2. By including leading zeros we can regard each number in S as a four-digit number. Partition S into the sets S_1', S_2', S_3', S_4' according to whether the 5 is in the first, second, third, or fourth position.

$$|S| = 4 \times (9 \times 9 \times 9) = 2,916.$$



Four Basic Counting Principles XIII

Example

How many different five-digit numbers can be constructed out of the digits 1, 1, 1, 3, 8?

Solution

1. The multiset is $\{3 \cdot 1, 1 \cdot 3, 1 \cdot 8\}$.
2. We have only two choices to make:
3. which position is to be occupied by the 3 (5 choices)
4. and then which position is to be occupied by the 8 (4 choices).
5. The remaining three places are occupied by 1's.
6. By the multiplication principle the answer is $5 \times 4 = 20$.
7. If the five digits are 1, 1, 1, 3, 3, the answer is 10, half as many.



Permutations of Sets I

- Let r be a positive integer. By an *r-permutation* of a set S of n elements we understand an ordered arrangement of r of the n elements.

Example

If $S = \{a, b, c\}$, then the 1, 2, and 3-permutations of S are

$$a \quad b \quad c,$$

$$ab \quad ac \quad ba \quad bc \quad ca \quad cb,$$

$$abc \quad acb \quad bac \quad bca \quad cab \quad cba,$$

respectively.

- We denote by $P(n, r)$ the number of r-permutations of an n-element set. If $r > n$, then $P(n, r) = 0$.



Permutations of Sets II

- An n-permutation of an n-element set S will be more simply called a *permutation of S* or a *permutation of n elements*.
  A permutation of a set S can be thought of as a listing of the elements of S in some order.

Theorem

For n and r positive integers with $r \leq n$,

$$P(n, r) = n \times (n-1) \times \cdots \times (n-r+1).$$

Definition

For a nonnegative integer n, We define $n!$ (read *n factorial*) by $n! = n \times (n-1) \times \cdots \times 1$, with the convention that $0! = 1$.

$$P(n, r) = \frac{n!}{(n-r)!}.$$



Permutations of Sets III

Example

The number of 4-letter "words" that can be formed by using each of the letters a, b, c, d, e at most once is $P(5, 4)$, and this equals $5!/(5-4)! = 120$. The number of 5-letter words equals $P(5, 5)$, which is also 120.

Example

The so-called "15 puzzle" consists of 15 sliding unit squares labeled with the numbers 1 through 15 and mounted in a 4-by-4 square frame as shown. The challenge of the puzzle is to move from the initial position shown to any specified position. By a position, we mean an arrangement of the 15 numbered squares in the frame with one empty unit square. What is the number of positions in the puzzle (ignoring whether it is possible to move to the position from the initial one)?

1	2	3	4
5	6	7	8
9	10	11	12
13	14	15	

15 puzzle



Permutations of Sets IV

Solution

$P(16, 15) = P(16, 16) = 16!$.

Example

What is the number of ways to order the 26 letters of the alphabet so that no two of the vowels a, e, i, o, and u occur consecutively?

Solution

1. Think of two main tasks.
2. The first is to decide how to order the permutations of the 21 consonants.
3. The second is to put the vowels in 5 of the 22 spaces before, between, and after the consonants.

$$21! \times P(22, 5) = 21! \times \frac{22!}{17!}.$$



Permutations of Sets V

Example

How many seven-digit numbers are there such that the digits are distinct integers taken from $\{1, 2, \cdots, 9\}$ and such that the digits 5 and 6 do not appear consecutively in either order?

Partition certain 7-permutations of the set $\{1, 2, \ldots, 9\}$ into 4 types:

1. neither 5 nor 6 appears as a digit, permutations of $\{1, 2, 3, 4, 7, 8, 9\}$ — $P(7, 7)$.
2. 5, but not 6 appears as a digit, with 5 being any one of the 7 digits and remaining 6 digits as a 6-permutation of $\{1, 2, 3, 4, 7, 8, 9\}$ — $7 \times P(7, 6)$.
3. 6, but not 5 appears as a digit, with 6 being any one of the 7 digits and remaining 6 digits as a 6-permutation of $\{1, 2, 3, 4, 7, 8, 9\}$ — $7 \times P(7, 6)$.
4. both 5 and 6 appear as digits.
    1. First digit equal to 5 and so second digit not equal to 6. There are 5 places for the 6 and the other 5 digits constitute a 5-permutation of $\{1, 2, 3, 4, 7, 8, 9\}$ — $5 \times P(7, 5)$.
    2. Last digit equal to 5 and so next to last digit not equal to 6 — $5 \times P(7, 5)$.
    3. A digit other than the first or last is equal to 5 There are 5 places for the 5 and then 4 places for the 6 — $5 \times 4 \times P(7, 5)$.



Permutations of Sets VI

Thus there are $2 \times 5 \times P(7,5) + 5 \times 4 \times P(7,5) = 30 \times P(7,5)$ numbers overall.

By the addition principle, the answer to the problem posed is $P(7,7) + 2 \times 7 \times P(7,6) + 30 \times P(7,5) = 151,200$.
An alternative and more easier way is to partition the entire collection T, of seven-digit numbers by distinct integers from $\{1,2,\ldots,9\}$, into two subsets S, which consists of those numbers in T in which 5 and 6 do not occur consecutively, and its complement $\overline{S}$ which consists of those numbers in T in which 5 and 6 do occur consecutively.

$$|S| = |T| - |\overline{S}| = P(9,7) - 2 \times 6 \times P(7,5) = 151,200.$$

- The permutations that we have just considered are more properly called *linear permutations*.
- If instead of arranging objects in a line, we arrange them in a circle — *circular permutation* – the number of permutations is smaller.



Permutations of Sets VII

$$\begin{array}{ccccccccccc} & 1 & & & 6 & & & 5 & & 4 & & 3 & & 2 & \\ 2 & & 6 & = & 1 & & 5 & = & 6 & & 4 & = & 5 & & 3 & = & 4 & & 2 & = & 3 & & 1 \\ 3 & & 5 & & 2 & & 4 & & 1 & & 3 & & 6 & & 2 & & 5 & & 1 & & 4 & & 6 \\ & 4 & & & & 3 & & & & 2 & & & & 1 & & & & 6 & & & & 5 & \\ 123456 & & & = & 234561 & & & = & 345612 & & & = & 456123 & & & = & 561234 & & & = & 612345 & & \end{array}$$

Theorem

The number of circular r-permutations of a set of n elements is given by

$$\frac{P(n,r)}{r} = \frac{n!}{r \cdot (n-r)!}.$$

In particular, the number of circular permutation of n elements is $(n-1)!$.

Example

Ten people, including two who do not wish to sit next to one another, are to be seated at a round table. How many circular seating arrangements are there?



Permutations of Sets VIII

Solution

$9! - 2 \times 8! = 7 \times 8!$ or $8 \times 7 \times 7! = 7 \times 8!$.

Example

The number of ways 12 different markings can be placed on a rotating drum is $P(12, 12)/12 = 11!$.

Example

What is the number of necklaces that can be made from 20 beads, each of a different color?

Solution

$19!/2$.



Combinations of Sets I

- Let r be a nonnegative integer. By an *r-combination* of a set S of n elements, we understand an unordered selection of r of the n objects of S.
- We denote by $\binom{n}{r}$ the number of r-combinations of an n-element set.

  Other common notations for these numbers are $C(n, r)$ and ${}_nC_r$.

Obviously,

$$\binom{n}{r} = 0 \qquad \text{if } r > n,$$

$$\binom{0}{r} = 0 \qquad \text{if } r > 0,$$

$$\binom{n}{0} = 1, \quad \binom{n}{1} = n, \quad \binom{n}{n} = 1, \quad \binom{0}{0} = 1.$$



Combinations of Sets II

Theorem

For $0 \leq r \leq n$,

$$P(n, r) = r! \binom{n}{r}$$

Hence,

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}.$$

Example

Twenty-five points are chosen in the plane so that no three of them are collinear. How many straight lines do they determine? How many triangles do they determine?



Combinations of Sets III

Solution

$\binom{25}{2} = \frac{25!}{2!23!} = 300$ and $\binom{25}{3} = \frac{25!}{3!22!} = 2300$.



Combinations of Sets IV

Example

There are 15 people enrolled in a mathematics course, but exactly 12 attend on any given day. The number of different ways that 12 students can be chosen is

$$\binom{15}{12} = \frac{15!}{12!3!}.$$

If there are 25 seats in the classroom, the 12 students could seat themselves in $P(25, 12) = 25!/13!$ ways. Thus, there are

$$\binom{15}{12} P(25, 12) = \frac{15!25!}{12!3!13!}$$

ways in which an instructor might see the 12 students in the classroom.



Combinations of Sets V

Example

How many 8-letter words can be constructed by using the 26 letters of the alphabet if each word contains 3, 4, or 5 vowels? It is understood that there is no restriction on the number of times a letter can be used in a word.

Solution

By the addition principle, the total number of words is

$$\binom{8}{3}5^3 21^5 + \binom{8}{4}5^4 21^4 + \binom{8}{5}5^5 21^3.$$

Corollary

For $0 \leq r \leq n$,

$$\binom{n}{r} = \binom{n}{n-r}.$$

Combinations of Sets VI

Theorem

We have

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n} = 2^n,$$

and the common value equals the number of combinations of an n-elements set.



Combinations of Sets VII

Proof

The number of combinations of an n-element set S is counted by either of the following ways.

1. Observe that every combination of S is an r-combination of S for some $r = 0, 1, 2, \ldots, n$. By the addition principle, it equals to

$$\binom{n}{0} + \binom{n}{1} + \binom{n}{2} + \cdots + \binom{n}{n},$$

2. In choosing a combination of S we have two choices to make for each of the n elements in S, either it goes into the combination or it doesn't. There are 2^n ways we can form a combination of S.

We now equate the two counts and complete the proof.



Combinations of Sets VIII

Example

The number of 2-combinations of the set $\{1, 2, \cdots, n\}$ of the first n positive integers is $\binom{n}{2}$. Partition the 2-combinations according to the largest integer they contain. For each $i = 1, 2, \cdots, n$, the number of 2-combinations in which i is the largest integer is $i - 1$ (the other integer can by any of $1, 2, \cdots, i - 1$). Equating the two counts, we obtain the identity

$$0 + 1 + 2 + \cdots + (n-1) = \binom{n}{2} = \frac{n(n-1)}{2}.$$



Permutations of Multisets I

- If S is a multiset, an *r-permutation* of S is an ordered arrangement of r of the objects of S.

Example

If $S = \{2 \cdot a, 1 \cdot b, 3 \cdot c\}$, then $acbc$ and $cbcc$ are 4-permutations of S, while $abccca$ is a permutation of S.

Theorem

Let S be a multiset with objects of k different types, where each has an infinite repetition number. Then the number of r-permutations of S is k^r.

Proof

In constructing an r-permutation of S, we can choose the first item to be an object of any one of the k types. Similarly, the second item can be an object of any one of the k types (since all repetition numbers of S are infinite), and so on. By the multiplication principle, the r items can be chosen in k^r ways.

Permutations of Multisets II

- The conclusion of the theorem remains true if the repetition numbers of the k different types of objects of S are all at least r.

Example

What is the number of ternary numerals with at most 4 digits? A *ternary numeral*, or base 3 numeral, is one arrived at by representing a number in terms of powers of 3. For instance, $46 = 1 \times 3^3 + 2 \times 3^2 + 0 \times 3^1 + 1 \times 3^0$, and so its ternary numeral is 1201.

Solution

The answer is the number of 4-permutations of the multiset $\{\infty \cdot 0, \infty \cdot 1, \infty \cdot 2\}$ or of the multiset $\{4 \cdot 0, 4 \cdot 1, 4 \cdot 2\}$, which equals $3^4 = 81$.



Permutations of Multisets III

Theorem

Let S be a multiset with objects of k different types with finite repetition numbers $n_1, n_2, \ldots, n_k$, respectively. Let the size of S be $n = n_1 + n_2 + \cdots + n_k$. Then the number of permutations of S equals

$$\frac{n!}{n_1!n_2!\cdots n_k!}.$$

1. We are given a multiset S having objects of k types, say $a_1, a_2, \ldots, a_k$, with repetition numbers $n_1, n_2, \ldots, n_k$, respectively, for a total of $n = n_1 + n_2 + \cdots + n_k$ objects.
2. There are n places, and we want to put exactly one of the objects of S in each of the places.
3. We first decide which places are to be occupied by the a_1's. There are n_1 a_1's in S, we must choose a subset of n_1 places from the set of n places, which can be done in $\binom{n}{n_1}$ ways.
4. We next decide which places are to be occupied by the a_2's. This can be done in $\binom{n-n_1}{n_2}$ ways.



Permutations of Multisets IV

5. Continuing like this, and we invoke the multiplication principle and find that the number of permutations of S equals

$$\begin{aligned}&\binom{n}{n_1}\binom{n-n_1}{n_2}\cdots\binom{n-n_1-\cdots-n_{k-1}}{n_k}\\&\quad=\frac{n!}{n_1!(n-n_1)!}\frac{(n-n_1)!}{n_2!(n-n_1-n_2)!}\cdots\frac{(n-n_1-\cdots-n_{k-1})!}{n_k!(n-n_1-\cdots-n_k)!},\end{aligned}$$

which, after cancellation, reduces to

$$\frac{n!}{n_1!n_2!\cdots n_k!0!}=\frac{n!}{n_1!n_2!\cdots n_k!}.$$

□



Permutations of Multisets V

Example

The number of permutations of the letters in the word MISSISSIPPI is $\frac{11!}{1!4!4!2!}$.
This equals the number of permutations of the multiset $\{1 \cdot M, 4 \cdot I, 4 \cdot S, 2 \cdot P\}$.

Example

Consider a set of the 4 objects $\{a, b, c, d\}$ that is to be partitioned into two sets, each of size 2. If the parts are not labeled, then there are 3 different partitions:

$$\{a, b\}, \{c, d\}; \qquad \{a, c\}, \{b, d\}; \qquad \{a, d\}, \{b, c\}.$$

However, if the parts are labeled, let us labeled them with colors red and blue. Then the number of partitions is greater; indeed, there are 6, since we can assign the labels red and blue to each part of a partition in 2 ways. For instance, for the partition $\{a, b\}, \{c, d\}$ we have

$$\boxed{\{a, b\}}, \boxed{\{c, d\}} \text{ and } \boxed{\{a, b\}}, \boxed{\{c, d\}}.$$

Permutations of Multisets VI

Theorem

Let n be a positive integer and let $n_1, n_2, \ldots, n_k$ be positive integers with $n = n_1 + n_2 + \cdots + n_k$. The number of ways to partition a set of n objects into k labeled boxes $B_1, B_2, \ldots, B_k$ in which Box 1 contains n_1 objects, Box 2 contains n_2 objects, ..., Box k contains n_k objects equals

$$\frac{n!}{n_1!n_2!\cdots n_k!}.$$

If the boxes are not labeled, the the number of partitions equals

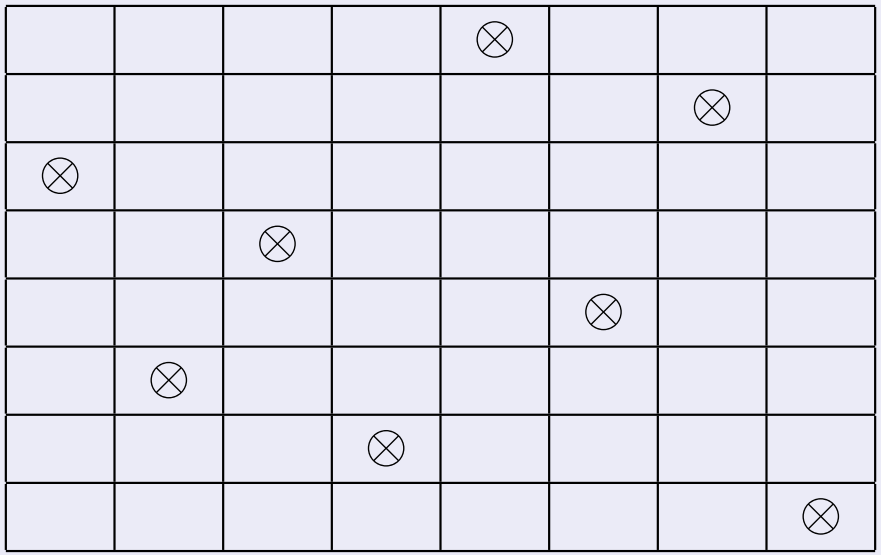
$$\frac{n!}{k!n_1!n_2!\cdots n_k!}.$$



Permutations of Multisets VII

Example

How many possibilities are there for 8 nonattacking rooks on an 8-by-8 chessboard?



An example of 8 nonattacking rooks on an 8-by-8 board

1. We give each square on the board a pair (i, j) of coordinates with i and j designating the row and column numbers of the square. Thus, i and j are integers between 1 and 8.

Permutations of Multisets VIII

2. There are must be exactly one rook in each row since there are to be 8 rooks on the board that cannot attack one another.
3. The rooks must occupy 8 squares with coordinates $(1, j_1), (2, j_2), \ldots, (8, j_8)$. But there must also be exactly one rook in each column so that no two of the numbers $j_1, j_2, \ldots, j_8$ can be equal. $j_1, j_2, \ldots, j_8$ must be a permutation of $\{1, 2, \ldots, 8\}$.
4. Conversely, if $j_1, j_2, \ldots, j_8$ is a permutation of $\{1, 2, \ldots, 8\}$, then putting rooks in the squares with coordinates $(1, j_1), (2, j_2), \ldots, (8, j_8)$, we arrive at 8 nonattacking rooks on the board.
5. Thus, we have a one-to-one correspondence between sets of 8 nonattacking rooks on the 8-by-8 board and permutation of $\{1, 2, \ldots, 8\}$. Therefore there are 8! ways to place 8 rooks on an 8-by-8 board so that they are nonattacking.

We implicitly assumed in the preceding argument that the rooks were *indistinguishable* from one another.

1. If we have 8 distinct rooks, say 8 rooks each colored with one of 8 different colors, then we have also to take into account which rook is in each of the 8 occupied squares.



Permutations of Multisets IX

2. As we look at the rooks from row 1 to row 8, we see a permutation of the 8 colors.
3. Thus, the number of ways to have 8 nonattacking rooks of 8 different colors on an 8-by-8 board equals $8!8! = (8!)^2$.

Now suppose that we have 1 red (R) rook, 3 blue (B) rooks, and 4 Yellow (Y) rooks. It is assumed that rooks of the same color are indistinguishable from one another. Now, as we look at the rooks from row 1 to row 8, we see a permutation of the colors of the multiset $\{1 \cdot R, 3 \cdot B, 4 \cdot Y\}$. The numbers of ways to place 1 red, 3 blue, and 4 yellow rooks on an 8-by-8 board so that no rook can attack another equals $8! \frac{8!}{1!3!4!} = \frac{(8!)^2}{1!3!4!}$.



Permutations of Multisets X

Theorem

There are n rooks of k colors with n_1 rooks of the first color, n_2 rooks of the second color, ..., and n_k rooks of the kth color. The number of ways to arrange these rooks on an n-by-n board so that no rook can attack another equals

$$n! \frac{n!}{n_1! n_2! \cdots n_k!} = \frac{(n!)^2}{n_1! n_2! \cdots n_k!}.$$

Example

Consider the multiset $S = \{3 \cdot a, 2 \cdot b, 4 \cdot c\}$ of 9 objects of 3 types. Find the number of 8-permutations of S.



Permutations of Multisets XI

Solution

8 permutations of S can be partitioned into permutations of $\{2 \cdot a, 2 \cdot b, 4 \cdot c\}$, $\{3 \cdot a, 1 \cdot b, 4 \cdot c\}$, and $\{3 \cdot a, 2 \cdot b, 3 \cdot c\}$, of which there are $\frac{8!}{2!2!4!} = 420$, $\frac{8!}{3!1!4!} = 280$, and $\frac{8!}{3!2!3!} = 560$. Thus, the number of 8-permutations of S is $420 + 280 + 560 = 1260$.



Combinations of Multisets I

- If S is a multiset, an *r-combination* of S is an unordered arrangement of r of the objects of S. Thus, an r-combination of S is itself a multiset, a *submultiset* of S.

Example

If $S = \{2 \cdot a, 1 \cdot b, 3 \cdot c\}$, then the 3-combinations of S are $\{2 \cdot a, 1 \cdot b\}$, $\{2 \cdot a, 1 \cdot c\}$, $\{1 \cdot a, 1 \cdot b, 1 \cdot c\}$, $\{1 \cdot a, 2 \cdot c\}$, $\{1 \cdot b, 2 \cdot c\}$, $\{3 \cdot c\}$.

Theorem

Let S be a multiset with objects of k different types, each with an infinite repetition number. Then the number of r-combinations of S equals

$$\binom{r+k-1}{r} = \binom{r+k-1}{k-1}.$$

Let the k types of objects of S be $a_1, a_2, \ldots, a_k$ so that $S = \{\infty \cdot a_1, \infty \cdot a_2, \ldots, \infty \cdot a_k\}$.



Combinations of Multisets II

1. Any r-combination of S is of the form $\{x_1 \cdot a_1, x_2 \cdot a_2, \ldots, x_k \cdot a_k\}$ where $x_1, x_2, \ldots, x_k$ are nonnegative integers with $x_1 + x_2 + \cdots + x_k = r$.
2. Conversely, every sequence $x_1, x_2, \ldots, x_k$ of nonnegative integers with $x_1 + x_2 + \cdots + x_k = r$ corresponds to an r-combination of S.
3. The number of r-combinations of S equals the number of solutions of the equation

$$x_1 + x_2 + \cdots + x_k = r,$$

where $x_1, x_2, \ldots, x_k$ are nonnegative integers.
4. We show that the number of these solutions equals the number of permutations of the multiset $T = \{r \cdot 1, (k-1) \cdot *\}$ of objects of two different types, equivalently, the number of sequences of $*$'s and 1's of length $r + k - 1$ in which there are r 1's and $k - 1$ $*$'s.
    1. Give a permutation of T, the $k - 1$ $*$'s divide the r 1's into k groups. Let there be x_1 1's to the left of the first $*$, x_2 1's between the first and the second $*$, ..., and x_k 1's to the right of the last $*$. Then $x_1, x_2, \ldots, x_k$ are nonnegative integers with $x_1 + x_2 + \cdots + x_k = r$.
    2. Conversely, given nonnegative integers $x_1, x_2, \ldots, x_k$ with $x_1 + x_2 + \cdots + x_k = r$, we can construct a permutation of T.



Combinations of Multisets III

Thus, the number of r-combinations of the multiset S equals $\frac{(r+k-1)!}{r!(k-1)!} = \binom{r+k-1}{r}$. □

Another way of phrasing the previous theorem

The number of r-combinations of k distinct objects, each available in unlimited supply, equals $\binom{r+k-1}{r}$.

- The theorem remains true if the repetition numbers of the k distinct objects of S are all at least r.

Example

A bakery boasts 8 varieties of doughnuts. If a box of doughnuts contains 1 dozen, how many different options are there for a box of doughnuts?

Solution

$\binom{12+8-1}{12} = \binom{19}{12}$.

Combinations of Multisets IV

Example

What is the number of nondecreasing sequences of length r whose terms are taken from $1, 2, \cdots, k$?

Solution

$\binom{r+k-1}{r}$.

Example

Let S be the multiset $\{10 \cdot a, 10 \cdot b, 10 \cdot c, 10 \cdot d\}$ with objects of four types, a, b, c and d. What is the number of 10-combinations of S which have the property that each of the four types of objects occurs at least once?



Combinations of Multisets V

1. The answer is the number of *positive* integral solutions of

$$x_1 + x_2 + x_3 + x_4 = 10,$$

where x_1, x_2, x_3, and x_4 represent the numbers of a, b, c, and d, respectively.
2. Since the repetition numbers all equal to 10, which is the size of the combinations being counted, we can ignore the repetition numbers. We then perform the changes of variables

$$y_1 = x_1 - 1, \quad y_2 = x_2 - 1, \quad y_3 = x_3 - 1, \quad y_4 = x_4 - 1$$

to get

$$y_1 + y_2 + y_3 + y_4 = 6,$$

where the y_i's are to be nonnegative.
3. The number of nonnegative integral solutions of the new equation is $\binom{6+4-1}{6} = \binom{9}{6} = 84$.



Combinations of Multisets VI

Example

Continuing with the doughnut example, we see that the number of different options for boxes of doughnuts containing at least one doughnut of each of the 8 varieties equals

$$\binom{12-8\times 1+8-1}{12-8\times 1}=\binom{11}{4}=330.$$

Example

What is the number of integral solutions of the equation

$$x_1+x_2+x_3+x_4=20,$$

in which

$$x_1\geq 3,\quad x_2\geq 1,\quad x_3\geq 0\quad \text{and}\quad x_4\geq 5?$$



Combinations of Multisets VII

Solution

$\binom{20-3-1-0-5+4-1}{20-3-1-0-5} = \binom{14}{11} = 364.$

- It is more difficult to count the number of r-combinations of a multiset

$$S = \{n_1 \cdot a_1, n_2 \cdot a_2, \ldots, n_k \cdot a_k\}$$

with k types of objects and general repetition numbers $n_1, n_2, \ldots, n_k$. The number of r-combinations of S is the same as the number of integral solutions of

$$x_1 + x_2 + \cdots + x_k = r$$

where

$$0 \le x_1 \le n_1, \quad 0 \le x_2 \le n_2, \quad \ldots, \quad 0 \le x_k \le n_k.$$

In the chapter of inclusion-exclusion principle, it will be provided a satisfactory method for this case.

